

NANDHAKUMAR G

MECHANICAL ENGINEERING UNDERGRADUATE (2027 BATCH)

Analytical and detail-oriented Mechanical Engineering student with hands-on industrial traineeship experience in heavy manufacturing machinery operations, continuous casting, and sheet metal design verification. Proficient in engineering drawing interpretation and parametric CAD modeling, with specialized focus on additive manufacturing and electromechanical control systems (ROS and sensor networks). Skilled in compiling structured technical reports, specifications sheets, and parts diagnostics protocols.

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TECHNICAL FOCUS

- Drawing & Schemes:** Strong capabilities in reading, analyzing, and verifying engineering blueprints and electrical schematics layouts.
- System Documentation:** Experienced in drafting structured mechanical specifications, parts manuals, assembly processes, and validation checklists.
- Additive Manufacturing:** Applied training in DFM guidelines for FDM/SLA 3D printing parameters, optimizing infills, and printing tolerances.
- Diagnostics Support:** Quick to troubleshoot electromechanical systems, control loops, and component assemblies clearances.

CORE COMPETENCIES

TECHNICAL SUPPORT & DOCS

- Drawing Reading
- Technical Writing
- Parts Manuals
- Component Diagrams

CAD PROTOTYPING & DFM

- 3D Assemblies
- Additive Manufacturing
- Tolerance Stacks
- DFM Principles
- Stress Analysis

TECHNICAL TOOLS & SOFTWARE

- Fusion 360
- SolidWorks
- Siemens NX
- AutoCAD
- Python
- C++
- MATLAB
- COMSOL

KEY ATTRIBUTES

- Analytical Problem Solving:** Capable of debugging electromechanical control loops and structural clearance conflicts.
- Technical Communication:** Adept at explaining complex physical parts layouts and assemblies in written specifications.
- Rapid Adaptation:** Demonstrates a quick learning curve when mastering new mechanical components and software tools.

PROFESSIONAL EXPERIENCE (INTERNSHIPS)

Reverse Engineering Intern Femlogic Industries, Chennai

June 2026

- Performed dimensional inspection and geometric data capture of legacy mechanical parts using precision metrology tools (calipers, micrometers, and thread gauges).
- Reconstructed physical components into fully editable parametric 3D assemblies, resolving design deviations and model geometry errors.
- Validated clearance configurations, fit parameters, and tolerance stacks to ensure compatibility with modern mechanical assemblies.
- Modified CAD files to optimize structures for additive manufacturing (3D printing), optimizing print orientation, tolerances, and support layouts.

Industrial Engineering Trainee Steel Authority of India Ltd. (SAIL), Salem Steel Plant

Dec 2025

- Inspected operations of integrated hot rolling steel mill structures, high-pressure hydraulics control systems, and heavy casting assemblies.
- Analyzed rolling mill load distributions and continuous casting roll alignment procedures, documenting thermal expansion tolerances.
- Read and interpreted complex mill blueprints, piping layouts, and structural drawings to verify machinery positioning.
- Documented preventive maintenance protocols, equipment operational schedules, and machinery safety checklists.
- Collaborated with senior plant technicians to evaluate machinery performance parameters, wear indicators, and component fatigue life.

Product Design Intern Femlogic Technologies Pvt. Ltd., Chennai

June 2025 - July 2025

- Contributed to the structural assembly design and spatial packaging of the Vande Bharat train passenger seating modules in CAD.
- Collaborated with senior engineers to apply Design for Manufacturing (DFM) and Design for Assembly (DFA) guidelines to thin sheet metal stampings.
- Drafted comprehensive parts interface reports, bills of materials (BOM), and structural verification documents for engineering teams.
- Assisted in conducting tolerance analyses, interference checks, and clearance verification on structural mounting plates.

Production Engineering Intern K.M. Knitwear Pvt. Ltd., Tirupur

July 2024 - August 2024

- Analyzed material flow diagrams of floor packaging machinery, identifying bottlenecks to boost packaging throughput rate by 10%.
- Drafted structural design layouts and dimensioned drawings for materials storage components to secure items during shipping cycles.
- Participated in mechanical troubleshooting, linkage adjustments, and diagnostic checks on machinery drive linkages and gearboxes.

EDUCATION

B.E. Mechanical Engineering St. Joseph's College of Engineering <i>Coursework: Machine Design, Fluid Mechanics, Control Theory</i>	2023 - Present
Class XII (Computer Science) Shri Swamy Matriculation School <i>Physics, Mathematics, Computer OOP</i>	2022 - 2023
Class X (Secondary School) Shri Swamy Matriculation School <i>Matriculation State Board Curriculum</i>	2020 - 2021

CREDENTIALS & CERTS

Autodesk Certified Fusion Designer Professional CAD Design
Certified Solidworks Designer Mechanical Assemblies & DFM Spec
Siemens NX CAD Designer Advanced Surfaces & Modeling Spec
AutoCAD for 2D Modelling Technical Drafting Standards
Certified in Python Advanced Computational Data Structures
C++ Programming OOP & Control Loops Logic

LANGUAGES

English Fluent	Tamil Native
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SELECTED ENGINEERING PROJECTS

Project EcoFloat Hackathon Winner & Shortlisted for Niral Regional Level SYSTEMS LEAD ROS · IoT · Fusion 360 · AM	- Designed and co-developed an autonomous twin-hull catamaran robot for floating plastic waste collection and real-time chemical water diagnostics. - Configured sensor node communications (serial ports interface) and path tracking scripts integrated with ROS edge nodes. - Designed the full physical parametric assembly in Fusion 360, selecting materials for structural load, center of mass, and buoyancy calculations. - Optimized components for additive manufacturing (3D printing), adjusting infill densities and geometries to survive water-current stresses. - Compiled all system technical reports, physical assembly diagrams, parts manuals, and standard operational documentation.
Hemispherical CPV Solar Tracking System MECHANICAL DESIGNER COMSOL · Solar Math · Fusion 360	- Modeled a compound dual-axis Concentrator Photovoltaic solar tracking dome structure designed to maximize solar vector energy absorption. - Conducted finite element (FEA) thermal and structural safety factor simulations in COMSOL to validate high wind-load tolerances. - Drafted detailed mechanical drawings, engineering blueprints, assembly calculations, and mathematical solar tracking formulas. - Designed custom linkages and mounting brackets, ensuring precise motor alignment and gear reduction ratios.
EV Regenerative Braking Control Loop DEVELOPER & RESEARCHER MATLAB Simulink · Powertrain Sim	- Designed an active kinetic-to-electrical energy recovery control loop for electric vehicle prototypes using MATLAB Simulink. - Integrated battery management system (BMS) controls to monitor thermal state-of-charge curves, extending battery lifecycle metrics by 20%. - Compiled comprehensive simulation test reports, transient response graphs, and component specifications registers. - Simulated regenerative braking performance under varying drive cycle profiles to optimize energy harvesting efficiency.